Individual Weekly Report – Week 8

Dashboard Construction

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Department	Artificial Intelligence and Data Science
Project Title	Youth Smoking and Drug Consumption: From Raw Data to Visual Insights
Role	Data Visualization and Dashboard Development
Week	Week 8 – Dashboard Construction

Objective of the Week

The aim of this week was to build the interactive dashboard. This was the main development task of my role and brought together the visualizations created in all the earlier weeks.

Work Done

Developing the interactive dashboard: I developed the dashboard using Streamlit and Plotly. Streamlit was used to build the web page, and Plotly was used to create the interactive charts that users can explore.

Integrating charts and graphs: I added the different charts and graphs created earlier into the dashboard, so that all the important results of the project can be seen in one place.

Designing dashboard sections: I divided the dashboard into sections so that users can move through the information step by step. This makes the dashboard easier to understand and use.

Improving layout and user interaction: I improved the layout and the way users interact with the dashboard, so that it looks neat and the charts respond easily to the user's actions.

Tools and Techniques Used

Streamlit for the dashboard interface and Plotly for interactive charts.

Outcome of the Week

A working interactive dashboard was completed, showing the results of the project in a clear and organized way.

What I Learned

I learned how to build an interactive dashboard with Streamlit and Plotly, and how important layout and user-friendliness are in a dashboard.

Contribution to the Project

The dashboard became the main output of the project and the way the findings are communicated to others.

Challenges Faced

Fitting many charts into one dashboard without making it crowded was the main challenge. I handled this by dividing the dashboard into sections and improving the layout step by step until it looked neat and was easy to use.

Plan for Next Week

In Week 9, I will demonstrate the completed dashboard, explain the charts and present the key insights during the final presentation and viva.

Weekly Summary

Week 8 was the most important week for my role. The interactive dashboard brought together all the visualizations of the project in one place.